Espaço Dual

E1 Sejam $f : \mathbb{K}^n \to \mathbb{K}$ funcional linear e $v_1, \dots, v_n$ base de $\mathbb{K}^n$. Escreve $x = \sum x_i v_i \in \mathbb{K}^n$ e aplica f, então tem-se $f(x) = f(\sum x_i v_i) = \sum x_i f(v_i)$. Define $f(v_i) \coloneqq \alpha_i$ e obtém-se o resultado. Reciprocamente, dados $x = \sum x_i v_i, y = \sum y_i v_i \in \mathbb{K}^n$, toma $f(x+y) = f\left(\sum x_i v_i + \sum y_i v_i\right) = f\left(\sum (x_i + y_i) v_i\right) = \sum (x_i + y_i) f(v_i) = \sum x_i \alpha_i + \sum y_i \alpha_i = f(x) + f(y)$. Ademais, $f(\lambda x) = f\left(\lambda \sum x_i v_i\right) = f\left(\sum (\lambda x_i) v_i\right) = \sum \lambda x_i f(v_i) = \lambda \sum x_i \alpha_i = \lambda f(x)$ para todo $\lambda \in \mathbb{K}$. □

E2 Sejam $h(x), p(x), q(x) \in \mathcal{P}_2(\mathbb{R})$ tal que $h(x) = p(x) + q(x)$. Então $f_\alpha = (p(x) + q(x)) = f_\alpha(h(x)) = h(\alpha) = p(\alpha) + q(\alpha) = f_\alpha(p(x)) + f_\alpha(q(x))$. Toma $h(x) = \lambda p(x), \lambda \in \mathbb{R}$ e faz $f_\alpha(\lambda p(x)) = f_\alpha(h(x)) = h(\alpha) = \lambda p(\alpha) = \lambda f_\alpha(p(x))$. □

E3 Escreve

$$c_1 f_1 + c_2 f_2 + c_3 f_3 = 0, \quad c_1, c_2, c_3 \in \mathbb{R} \tag{1}$$
$$(c_1 f_1 + c_2 f_2 + c_3 f_3)(p(x)) = 0, \quad \forall p(x) \in \mathcal{P}_2(\mathbb{R}) \tag{2}$$
$$c_1 f_1(p(x)) + c_2 f_2(p(x)) + c_3 f_3(p(x)) = 0 \tag{3}$$
$$c_1 p(0) + c_2 p(1) + c_3 p(2) = 0 \tag{4}$$

toma $p(x) = (x-2)(x-3)$ e vem $c_1 = 0$. Faz $p(x) = (x-1)(x-3)$ e obtém $c_2 = 0$. Finalmente, com $p(x) = (x-1)(x-2)$ tem-se $c_3 = 0$. Logo, f_1, f_2 e f_3 são linearmente independentes. □

E4 (a) Nota que $\dim \mathcal{P}_2(\mathbb{R}) = 3$, então basta mostrar que f_1, f_2, f_3 é LI. Escreve

$$c_1 f_1 + c_2 f_2 + c_3 f_3 = 0, \quad c_1, c_2, c_3 \in \mathbb{R} \tag{5}$$
$$(c_1 f_1 + c_2 f_2 + c_3 f_3)(p(x)) = 0, \quad \forall p(x) \in \mathcal{P}_2(\mathbb{R}) \tag{6}$$
$$c_1 f_1(p(x)) + c_2 f_2(p(x)) + c_3 f_3(p(x)) = 0 \tag{7}$$
$$c_1 \int_0^1 p(x)\,dx + c_2 \int_0^2 p(x)\,dx + c_3 \int_0^3 p(x)\,dx = 0. \tag{8}$$

Toma $p(x) = 1$, $p(x) = 2x$ e $p(x) = 3x^2$ e obtém o sistema linear

$$\begin{cases} c_1 + 2c_2 + 3c_3 = 0 \\ c_1 + 4c_2 + 9c_3 = 0 \\ c_1 + 8c_2 + 27c_3 = 0 \end{cases}$$

cuja única solução é $c_1 = c_2 = c_3 = 0$. Logo, f_1, f_2, f_3 são linearmente independentes e formam uma base de $(\mathcal{P}_2(\mathbb{R}))^*$.

(b) Seja p_1, p_2, p_3 base de $\mathcal{P}_2(\mathbb{R})$ da qual f_1, f_2, f_3 é base dual. Então, $f_i(p_j) = \delta_{ij}$, onde δ_{ij} é o delta de Kronecker. Escreve $p_j(x) = a_j x^2 + b_j x + c_j$ e resolve o sistema

$$\begin{cases} f_1(p_j) = \int_0^1 (a_j x^2 + b_j x + c_j)\,dx = \frac{a_j}{3} + \frac{b_j}{2} + c_j = \delta_{1j} \\ f_2(p_j) = \int_0^2 (a_j x^2 + b_j x + c_j)\,dx = \frac{8a_j}{3} + 2b_j + 2c_j = \delta_{2j} \\ f_3(p_j) = \int_0^3 (a_j x^2 + b_j x + c_j)\,dx = 9a_j + \frac{9b_j}{2} + 3c_j = \delta_{3j} \end{cases}$$

para $j = 1, 2, 3$. Obtém-se $p_1(x) = 3 - 5x + \frac{3}{2}x^2$, $p_2(x) = -\frac{3}{2} + 4x - \frac{3}{2}x^2$ e $p_3(x) = \frac{1}{3} + x - \frac{1}{2}x^2$.

E5 (a) Para $u = (a, b, c) \in \mathbb{R}^3$, escreve

$$f(a,b,c) = f(\alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3), \quad \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R} \tag{9}$$
$$= \alpha_1 f(u_1) + \alpha_2 f(u_2) + \alpha_3 f(u_3) \tag{10}$$
$$= \alpha_1 - \alpha_2 + 3\alpha_3. \tag{11}$$

(b) Encontra os coeficientes $\beta_1, \beta_2, \beta_3$ tais que $(2,3,-1) = \beta_1 u_1 + \beta_2 u_2 + \beta_3 u_3 = \beta_1(1,0,1) + \beta_2(0,1,-2) + \beta_3(-1,1,0)$. Resolve o sistema

$$\begin{cases} \beta_1 - \beta_3 = 2 \\ \beta_2 + \beta_3 = 3 \\ \beta_1 - 2\beta_2 = -1 \end{cases}$$

e obtém $\beta_1 = 3$, $\beta_2 = 2$ e $\beta_3 = 1$. Logo, $f(2,3,-1) = \beta_1 f(u_1) + \beta_2 f(u_2) + \beta_3 f(u_3) = 0 + 0 + \beta_3 f(u_3) \neq 0$.

E6 Seja $\beta^* = \{f_1, f_2, f_3\}$ base dual de $\beta = \{v_1, v_2, v_3\}$, onde $v_1 = (1,0,-1)$, $v_2 = (1,1,1)$ e $v_3 = (2,2,0)$. Para qualquer $(a,b,c) \in \mathbb{C}^3$, tem-se $(a,b,c) = \alpha_1(1,0,-1) + \alpha_2(1,1,1) + \alpha_3(2,2,0)$. Resolve o sistema

$$\begin{cases} \alpha_1 + \alpha_2 + 2\alpha_3 = a \\ \alpha_2 + 2\alpha_3 = b \\ -\alpha_1 + \alpha_2 = c \end{cases}$$

e obtém $f_1(a,b,c) = \alpha_1 = \frac{a+b-2c}{3}$, $f_2(a,b,c) = \alpha_2 = \frac{a+b+c}{3}$ e $f_3(a,b,c) = \alpha_3 = \frac{a-2b+c}{6}$.

E7 Seja $p_1(x) = a_0 + a_1 x + a_2 x^2$, $p_2(x) = b_0 + b_1 x + b_2 x^2$ e $p_3(x) = c_0 + c_1 x + c_2 x^2$ base de $\mathcal{P}_2(\mathbb{R})$ tal que $\phi_i(p_j) = \delta_{ij}$, onde ϕ_1, ϕ_2, ϕ_3 é a base dual dada. Resolve o sistema

$$\begin{cases} \phi_1(p_j) = \int_0^1 (a_0 + a_1 + a_2)\, dx = \delta_{1j} \\ \phi_2(p_j) = (a_0 + a_1 x + a_2 x^2)'(1) = \delta_{2j} \\ \phi_3(p_j) = a_0 = \delta_{3j} \end{cases}$$

para $j = 1, 2, 3$. Obtém-se $p_1(x) = 3x - \frac{3}{2}x^2$, $p_2(x) = -\frac{1}{2}x + \frac{3}{4}x^2$ e $p_3(x) = 1 - 2x$.